Post-selection inference in multiverse meta-analysis

Matteo Manente[1], Livio Finos[2], Gianmarco Altoè[1], and Filippo Gambarota[1]

[1]Department of Developmental Psychology and Socialization, University of Padova

[2]Department of Statistical Sciences, University of Padova

Author Note

Correspondence concerning this article should be addressed to Filippo Gambarota, Department of Developmental Psychology and Socialization, University of Padova, Via Venezia 8, Padova, Italy 35131, Email: filippo.gambarota@unipd.it

Abstract

Multiverse analyses evaluate how research conclusions vary across multiple reasonable analytical specifications, but formal inference across the resulting set of dependent tests remains challenging. This problem is particularly relevant in meta-analysis, where inferential uncertainty arises both within individual meta-analytic models and across alternative analytical specifications. We introduce Post-Selection Inference in Multiverse Meta-Analysis (PIMMA), an extension of the PIMA framework to mixed-effects meta-analysis and meta-regression. The method derives study-level sign-flipping score contributions for specification-level inference and applies common sign-flipping transformations across specifications to obtain global and multiplicity-adjusted post-selection inference while accounting for dependence among overlapping meta-analytic models. In a simulation study, the specification-level sign-flipping procedure showed adequate Type I error control and competitive power across the conditions considered. We illustrate PIMMA using a multiverse meta-analysis of cognitive behavioral therapy for depression, comprising 438 admissible specifications based on different evidence-selection and modeling decisions. The global null hypothesis was rejected, and 378 specifications (86.3%) remained significant after single-step max-T adjustment, whereas none remained significant after Holm correction applied to the same specification-level tests. PIMMA therefore provides an inferential complement to descriptive multiverse meta-analysis by allowing researchers to assess both whether evidence exists somewhere in the multiverse and which specification-specific conclusions remain supportable after accounting for multiplicity and dependence.

Keywords: template, demo

Post-selection inference in multiverse meta-analysis

Introduction

Researchers make many decisions throughout a study, including choices about measurement, data preprocessing, and statistical models. When several alternatives are reasonable and defensible, these choices constitute *researcher degrees of freedom* (Simmons et al., 2011). Such flexibility is common in psychology and the social sciences, but also occurs in fields such as medicine and biology (Errington et al., 2021; Nosek et al., 2022). Because different analytical choices can lead to different conclusions even when applied to the same data and research question (Silberzahn et al., 2018), researchers may face a multiverse of reasonable analytical specifications (Steegen et al., 2016).

Researcher degrees of freedom are not inherently problematic when analytical choices are determined independently of the observed results. However, when multiple reasonable specifications are examined and the results influence which analyses are emphasized or interpreted, a multiple-testing and selective-inference problem arises. Without appropriate adjustment, this can inflate the probability of false positive conclusions (Gelman & Loken, 2013; Wicherts et al., 2016).

Because researcher degrees of freedom are often unavoidable, the relevant issue is not their existence but whether their consequences are made transparent. *Multiverse analysis* addresses this problem by applying all reasonable combinations of analytical choices and jointly reporting the resulting estimates (Simonsohn et al., 2020; Steegen et al., 2016). This makes the variability induced by analytical decisions explicit and allows researchers to assess the robustness of their conclusions.

Although multiverse analysis was developed mainly for primary studies, meta-analyses involve many of the same researcher degrees of freedom. Meta-analysts must decide, for example, which studies to include, which effect-size metric and sampling variance to use, and which statistical model to fit. Additional choices arise with complex data structures requiring aggregation, imputation, or multilevel and multivariate modeling (Cheung, 2013; Van den

Noortgate et al., 2013). Different defensible choices can therefore produce different meta-analytic conclusions.

Assessing robustness to such choices is particularly important in meta-analysis because systematic reviews and meta-analyses can strongly influence scientific, clinical, and policy conclusions (Higgins, 2019; Murad et al., 2016). Accordingly, multiverse meta-analysis has recently been used to examine topics including psychotherapy and digital interventions for depression (Plessen et al., 2023, 2025), IQ and mental illness (Fries et al., 2025), oxytocin administration (Kang et al., 2025), and the effects of music listening on epilepsy (Oberleiter & Pietschnig, 2023).

Most multiverse methods remain primarily descriptive. Approaches such as the original multiverse analysis (Steegen et al., 2016) and the *Vibration of Effects* framework (Patel et al., 2015) summarize how estimates and p values vary across analytical specifications, but do not provide formal inference accounting for the fact that many specifications have been examined. From an inferential perspective, a multiverse is therefore a multiple-testing problem: many generally dependent hypotheses are evaluated on the same data, and interpreting their unadjusted p values can lead to misleading conclusions (Girardi et al., 2024; Goeman & Solari, 2014; Westfall & Young, 1992).

Inference across a multiverse can be performed globally or at the level of individual specifications. A global test evaluates the intersection null hypothesis that the effect is null in every specification; its rejection implies that at least one specification has a non-null effect but does not identify which one. This limitation can be substantial when hundreds or thousands of specifications are considered (Oberauer & Lewandowsky, 2019, p. 1609). In contrast, making inferential claims about individual specifications after inspecting the multiverse requires multiplicity-adjusted or post-selection inference.

Existing inferential multiverse methods have focused mainly on global inference. *Specification Curve Analysis* (SCA) (Simonsohn et al., 2020), which has also been applied to multiverse meta-analysis (e.g., Pietschnig et al., 2022; Plessen et al., 2023), uses resampling under

the null to compare the observed specification curve with its joint null distribution. However, its inferential procedure is primarily developed for linear-model settings and does not provide a general framework for post-selection inference on individual specifications.

The more recent *Post-selection Inference in Multiverse Analysis* (PIMA) framework (Girardi et al., 2024) addresses this limitation using sign-flipping score tests and joint resampling across specifications. PIMA provides both a global test and specification-specific multiplicity-adjusted inference with strong control of the family-wise error rate. Developed within the generalized linear model framework, it also applies more broadly than SCA across outcome distributions and model specifications.

Extending PIMA to meta-analysis is not straightforward because multiverse meta-analysis combines two inferential problems. First, inference *within* each specification may be poorly calibrated: standard Wald-type inference in random- and mixed-effects meta-analysis relies on large-sample approximations that can perform poorly with few studies (Follmann & Proschan, 1999; Viechtbauer et al., 2015). Second, inference *across* specifications creates a multiple-testing and post-selection problem among highly dependent tests. Addressing either problem alone is insufficient: valid specification-level tests still require multiplicity adjustment across the multiverse, while multiplicity correction cannot compensate for poorly calibrated individual tests.

Permutation-based inference has been proposed for several meta-analytic settings, including testing average effects in random-effects meta-analysis (Follmann & Proschan, 1999), moderators in meta-regression (Higgins & Thompson, 2004; Viechtbauer et al., 2015), and effects in multivariate random-effects meta-analysis (Noma et al., 2020). These procedures, however, were developed for inference within individual meta-analytic models rather than across a multiverse of specifications.

Score-based sign flipping provides a natural connection between these two inferential levels. Rather than permuting outcomes or moderator values, these methods resample study- or observation-level score contributions, which have mean zero under the null after appropriate nuisance adjustment (De Santis et al., 2025a; Hemerik et al., 2020). The same sign-flipping logic

can then be extended to simultaneous inference across many correlated models through joint resampling (De Santis et al., 2025b).

In this paper, we extend score-based sign-flipping inference and the PIMA framework to mixed-effects meta-analysis and meta-regression, introducing Post-Selection Inference in Multiverse Meta-Analysis (PIMMA). We first derive study-level score contributions for testing meta-analytic regression coefficients in the presence of nuisance parameters and between-study heterogeneity. We then show how common sign-flipping transformations across specifications provide global and specification-specific multiplicity-adjusted inference. We evaluate the specification-level sign-flipping procedure in a simulation study and illustrate PIMMA using a multiverse meta-analysis of Cognitive Behavioral Therapy (CBT) for depression based on the data of Plessen et al. (2023).

Mixed-Effects Meta-Regression

In this section, we present a general formulation of the mixed-effects meta-regression model, following the model formulation of Viechtbauer et al. (2015). In a meta-analysis with k independent studies, let y_i and v_i denote respectively the observed effect size estimate and the sampling variance for the i-th study. The model can be written as

$$\mathbf{y} = \mathbf{D}\boldsymbol{\beta} + \mathbf{u} + \mathbf{e},$$
$$\mathbf{u} \sim \mathcal{N}(\mathbf{0}, \tau^2\mathbf{I}),$$
$$\mathbf{e} \sim \mathcal{N}(\mathbf{0}, \mathbf{V}),$$

{#eq-meta-reg}

where $\mathbf{y}$ is the $k \times 1$ vector of observed effect size estimates, $\mathbf{D}$ is the $k \times p$ model matrix (including the intercept), and $\boldsymbol{\beta}$ is the vector of p regression coefficients. The vector $\mathbf{u}$ contains the study-level random effects and is assumed to follow a normal distribution with mean zero and variance τ^2, whereas $\mathbf{e}$ contains the within-study sampling errors. The sampling errors are assumed to be normally distributed with a diagonal variance-covariance matrix $\mathbf{V} = \text{diag}(v_1, \ldots, v_k)$, where the sampling variances v_i are treated as known. It follows that the

marginal distribution of the observed effect size estimates is

$$\mathbf{y} \sim \mathcal{N}\left(\mathbf{D}\boldsymbol{\beta}, \boldsymbol{\Sigma}\right), \qquad \boldsymbol{\Sigma} = \mathbf{V} + \tau^2\mathbf{I}.$$

{#eq-marginal}

For a fixed value of τ^2, the regression coefficients can be estimated by weighted least squares. Let $\mathbf{W} = \boldsymbol{\Sigma}^{-1} = (\mathbf{V} + \tau^2\mathbf{I})^{-1}$. The estimator of $\boldsymbol{\beta}$ is then

$$\mathbf{b} = (\mathbf{D}^\top\mathbf{W}\mathbf{D})^{-1}\mathbf{D}^\top\mathbf{W}\mathbf{y}.$$

{#eq-wls}

Since τ^2 is unknown, the weights used in practice are obtained by replacing τ^2 with an estimate $\hat{\tau}^2$, so that $\hat{\mathbf{W}} = (\mathbf{V} + \hat{\tau}^2\mathbf{I})^{-1}$. Several estimators have been proposed for τ^2 (see Viechtbauer, 2005). We focus here on the DerSimonian–Laird estimator (1986), as a commonly used non-iterative method-of-moments estimator, and on the restricted maximum likelihood (REML) method.

Both estimators can be expressed using the matrix

$$\mathbf{P} = \mathbf{W} - \mathbf{W}\mathbf{D}(\mathbf{D}^\top\mathbf{W}\mathbf{D})^{-1}\mathbf{D}^\top\mathbf{W}.$$

{#eq-het-p}

For the DerSimonian–Laird estimator, $\mathbf{W}$ is computed using only the inverse sampling variances, that is, $w_i = v_i^{-1}$. Since p denotes the total number of regression coefficients, including the intercept, the estimator is given by

$$\hat{\tau}^2_{\text{DL}} = \frac{\mathbf{y}^\top\mathbf{P}\mathbf{y} - (k - p)}{\text{tr}(\mathbf{P})},$$

{#eq-het-dl}

where negative $\hat{\tau}^2_{\text{DL}}$ values are truncated to zero.

Likelihood-based estimators are obtained from the marginal normal model in (**?@eq-marginal**). The log-likelihood of the random-effects model can be written as

$$\ell_{\text{ML}}(\boldsymbol{\beta}, \tau^2) = -\frac{1}{2}\left[k\log(2\pi) + \log|\boldsymbol{\Sigma}| + (\mathbf{y} - \mathbf{D}\boldsymbol{\beta})^\top \boldsymbol{\Sigma}^{-1}(\mathbf{y} - \mathbf{D}\boldsymbol{\beta})\right].$$

{#eq-ml-loglik}

For a fixed value of τ^2, maximizing (**?@eq-ml-loglik**) with respect to $\boldsymbol{\beta}$ gives the weighted least squares estimator in (**?@eq-wls**). Substituting this estimator into the likelihood yields the profiled log-likelihood

$$\ell^p_{\text{ML}}(\tau^2) = -\frac{1}{2}\left[k\log(2\pi) + \log|\boldsymbol{\Sigma}| + \mathbf{y}^\top \mathbf{P}\mathbf{y}\right].$$

{#eq-ml-profile}

However, maximum likelihood estimators of variance components are generally downward biased because they do not account for the loss of degrees of freedom associated with estimating the fixed effects. Restricted maximum likelihood addresses this issue by maximizing a likelihood based on linear combinations of the data that remove the fixed effects. Up to an additive constant independent of τ^2, the restricted log-likelihood can be written as

$$\ell_{\text{REML}}(\tau^2) \propto -\frac{1}{2}\left[\log|\boldsymbol{\Sigma}| + \log|\mathbf{D}^\top \mathbf{W}\mathbf{D}| + \mathbf{y}^\top \mathbf{P}\mathbf{y}\right].$$

{#eq-reml-loglik}

After $\hat{\tau}^2_{\text{REML}}$ has been obtained, the final estimates of the meta-regression coefficients are computed using $\hat{\tau}^2_{\text{REML}}$ in (**?@eq-wls**).

Several methods have been proposed to perform statistical inference on the regression coefficients in mixed-effects meta-regression models (see Viechtbauer et al., 2015). Here, we consider the three main approaches that will be compared in the simulation study: Wald-type inference, the Knapp–Hartung method (Knapp & Hartung, 2003), and permutations (Follmann & Proschan, 1999; Higgins & Thompson, 2004).

Under the standard Wald-type approach, the variance-covariance matrix of the estimated regression coefficients is estimated as

$$\hat{\mathbf{V}}_{\mathbf{b}} = (\mathbf{D}^{\top}\hat{\mathbf{W}}\mathbf{D})^{-1},$$

{#eq-vb-wald}

where

$$\hat{\mathbf{W}} = (\mathbf{V} + \hat{\tau}^2\mathbf{I})^{-1}.$$

The standard error of b_j is given by the square root of the corresponding diagonal element of $\hat{\mathbf{V}}_{\mathbf{b}}$. The Wald-type test statistic is then defined as

$$z_j = \frac{b_j}{\sqrt{\widehat{\mathrm{Var}}(b_j)}}.$$

{#eq-wald-stat}

Under the null hypothesis $H_0 : \beta_j = 0$, z_j is compared against the critical value calculated at the α level of a standard normal distribution.

The Knapp–Hartung method uses the same coefficient estimates but replaces the standard variance-covariance matrix (**?@eq-vb-wald**) with an adjusted version as $\hat{\mathbf{V}}_{\mathbf{b}\mathrm{KH}} = s^2\hat{\mathbf{V}}_{\mathbf{b}}$ with s^2 calculated as

$$s^2 = \frac{\mathbf{y}^{\top}\hat{\mathbf{P}}\mathbf{y}}{k - p}.$$

{#eq-kh-scale}

The test statistic is computed as in (**?@eq-wald-stat**), using the corresponding adjusted standard error, and is compared against a t distribution with $k - p$ degrees of freedom. A truncated version of the Knapp–Hartung method sets $s^2 = 1$ whenever $s^2 < 1$, thereby preventing the adjusted standard error from being smaller than the standard Wald-type standard error.

Permutation testing provides a nonparametric alternative for testing moderator effects. Following the implementation in `metafor::permutest()` and the permutation approach discussed by Viechtbauer et al. (2015), for models with moderators the rows of the complete model matrix $\mathbf{D}$ are permuted while the observed effect-size estimates and sampling variances are kept fixed. For each permutation, the complete mixed-effects meta-regression model is refitted, including re-estimation of the regression coefficients and τ^2. For a given coefficient b_j, let $z_j^{(1)}$ denote the Wald-type statistic obtained from the original data and let $z_j^{(r)}$, $r = 2, \ldots, B$, denote the statistics obtained from $B - 1$ random permutations. The original data configuration is included as the first configuration. For a two-sided test, the permutation p value is computed as

$$p_{\text{perm}} = \frac{\sum_{r=1}^{B} \mathbb{I}\left(|z_j^{(r)}| \geq |z_j^{(1)}|\right)}{B}.$$

{#eq-perm-p}

When all unique permutations of the model matrix are enumerated, this gives the exact permutation distribution. Otherwise, the distribution is approximated using a finite number of random permutations.

Meta-regression using flipscores

The mixed-effects meta-regression model can be reformulated as a score-based test. This provides the link between moderator testing in meta-analysis and the sign-flipping score framework used by PIMA. The key change is that resampling is no longer applied to the rows of the model matrix, as in the permutation test described above, but to the study-level contributions to the score.

We focus on a single focal regression coefficient. We partition the complete design matrix as $\mathbf{D} = [\mathbf{X}, \mathbf{Z}]$, where $\mathbf{X}$ is the $k \times 1$ design vector for the focal predictor and $\mathbf{Z}$ contains the nuisance predictors, including the intercept when appropriate. The corresponding model can be written as

$$\mathbf{y} \sim \mathcal{N}\left(\mathbf{X}\beta + \mathbf{Z}\boldsymbol{\gamma}, \boldsymbol{\Sigma}\right), \qquad \boldsymbol{\Sigma} = \mathbf{V} + \tau^2\mathbf{I}.$$

{#eq-flipmeta-model}

We test $H_0 : \beta = \beta_0$, usually with $\beta_0 = 0$, while $\boldsymbol{\gamma}$ and τ^2 remain nuisance parameters. Under the null model, we estimate $\hat{\boldsymbol{\gamma}}_0$ and $\hat{\tau}_0^2$ and define

$$\widehat{\boldsymbol{\Sigma}}_0 = \mathbf{V} + \hat{\tau}_0^2\mathbf{I}, \qquad \widehat{\mathbf{W}}_0 = \widehat{\boldsymbol{\Sigma}}_0^{-1}.$$

{#eq-null-weights}

The fitted values and residuals under the null are

$$\hat{\boldsymbol{\mu}}_0 = \mathbf{X}\beta_0 + \mathbf{Z}\hat{\boldsymbol{\gamma}}_0, \qquad \hat{\mathbf{r}}_0 = \mathbf{y} - \hat{\boldsymbol{\mu}}_0.$$

{#eq-null-residuals}

Although τ^2 is a nuisance parameter, it does not require an additional projection in the effective score. In the Gaussian meta-regression model, the parameters of the mean and the variance parameter τ^2 are Fisher-orthogonal,

$$\mathcal{I}_{\beta,\tau^2} = 0, \qquad \mathcal{I}_{\boldsymbol{\gamma},\tau^2} = \mathbf{0}.$$

This follows because the score for the mean parameters is linear in the residuals, whereas the score for τ^2 is quadratic in the residuals, and the corresponding cross-information is zero under the Gaussian model. Therefore, nuisance adjustment in the effective score is required for $\boldsymbol{\gamma}$, while τ^2 enters the test through the null working weights $\widehat{\mathbf{W}}_0$.

The connection with the general sign-flipping score framework follows directly from the Gaussian identity-link form of the meta-regression model. In the general framework, let $\mathbf{V}_Y = \mathrm{Var}(\mathbf{y})$ denote the response variance matrix, to distinguish it from the within-study sampling-variance matrix $\mathbf{V}$ used in meta-analysis. The working weight matrix is

$$\mathbf{W} = \mathbf{D}_\mu \mathbf{V}_Y^{-1} \mathbf{D}_\mu,$$

where $\mathbf{D}_\mu = \operatorname{diag}(\partial \mu_i / \partial \eta_i)$. For the Gaussian identity-link meta-regression model, $\mathbf{D}_\mu = \mathbf{I}$ and

$$\mathbf{V}_Y = \mathbf{\Sigma} = \mathbf{V} + \tau^2 \mathbf{I}.$$

Therefore,

$$\mathbf{W} = \mathbf{\Sigma}^{-1}, \qquad \mathbf{V}_Y^{-1/2} = \mathbf{W}^{1/2}.$$

The general effective score of Hemerik et al. (2020) and De Santis et al. (2025a) therefore reduces directly to the following score for mixed-effects meta-regression. Define the weighted hat matrix for the nuisance design as

$$\mathbf{H}_Z = \widehat{\mathbf{W}}_0^{1/2} \mathbf{Z} \left(\mathbf{Z}^\top \widehat{\mathbf{W}}_0 \mathbf{Z} \right)^{-1} \mathbf{Z}^\top \widehat{\mathbf{W}}_0^{1/2}.$$

{#eq-hz}

The effective score, evaluated under H_0, is then

$$S_{\text{eff}} = k^{-1/2} \mathbf{X}^\top \widehat{\mathbf{W}}_0^{1/2} (\mathbf{I} - \mathbf{H}_Z) \widehat{\mathbf{W}}_0^{1/2} \hat{\mathbf{r}}_0.$$

{#eq-effective-score-observed}

This is the direct specialization of the general flipscores effective score to the marginal Gaussian meta-regression model. Equivalently, it can be written as a sum of study-level effective score contributions,

$$S_{\text{eff}} = k^{-1/2} \sum_{i=1}^{k} \nu_i,$$

{#eq-score-contributions-observed}

where ν_i is the contribution of study i to the effective score. Define the focal predictor residualized with respect to the nuisance predictors as

$$\widetilde{\mathbf{X}} = \mathbf{X} - \mathbf{Z}\left(\mathbf{Z}^\top \widehat{\mathbf{W}}_0 \mathbf{Z}\right)^{-1} \mathbf{Z}^\top \widehat{\mathbf{W}}_0 \mathbf{X}.$$

The contribution of study i can then be written as

$$\nu_i = \tilde{x}_i \hat{w}_{0i} \hat{r}_{0i}.$$

{#eq-score-contribution-general}

With a single focal moderator and an intercept-only nuisance model, this reduces to

$$\nu_i = (x_i - \bar{x}_w)\hat{w}_{0i}\hat{r}_{0i}, \qquad \bar{x}_w = \frac{\sum_{i=1}^k \hat{w}_{0i} x_i}{\sum_{i=1}^k \hat{w}_{0i}}.$$

{#eq-simple-contribution}

Thus, in this case, the effective score is a weighted association between the centered moderator and the residuals under the null model. Under H_0, the ideal effective score contributions based on the true nuisance quantities have mean zero. Replacing the nuisance quantities by their estimates under the null yields the asymptotically equivalent score used in practice (De Santis et al., 2025a; Hemerik et al., 2020).

Let $\mathbf{F}^{(r)} = \mathrm{diag}(g_1^{(r)}, \ldots, g_k^{(r)})$ be a diagonal sign-flipping matrix, with $g_i^{(r)} \in \{-1, +1\}$ and $r = 1, \ldots, B$. The first flip is fixed as the identity, $\mathbf{F}^{(1)} = \mathbf{I}$, and the remaining $B - 1$ matrices are sampled at random. For flip r, the flipped effective score is

$$S_{\text{eff}}^{(r)} = k^{-1/2} \mathbf{X}^\top \widehat{\mathbf{W}}_0^{1/2} (\mathbf{I} - \mathbf{H}_Z) \widehat{\mathbf{W}}_0^{1/2} \mathbf{F}^{(r)} \hat{\mathbf{r}}_0,$$

{#eq-effective-score-flipped}

or, equivalently,

$$S_{\text{eff}}^{(r)} = k^{-1/2} \sum_{i=1}^{k} g_i^{(r)} v_i.$$

{#eq-score-contributions-flipped}

Following the standardized sign-flip score test, each flipped score is standardized using the leading term of its flip-specific conditional variance (De Santis et al., 2025a; Girardi et al., 2024). In the present meta-regression setting, this term is estimated as

$$\widehat{C}^{(r)} = k^{-1} \mathbf{X}^\top \widehat{\mathbf{W}}_0^{1/2} (\mathbf{I} - \mathbf{H}_Z) \mathbf{F}^{(r)} (\mathbf{I} - \mathbf{H}_Z) \mathbf{F}^{(r)} (\mathbf{I} - \mathbf{H}_Z) \widehat{\mathbf{W}}_0^{1/2} \mathbf{X}.$$

{#eq-flip-variance}

The standardized two-sided statistic is therefore

$$T^{(r)} = \left| \frac{S_{\text{eff}}^{(r)}}{\sqrt{\widehat{C}^{(r)}}} \right|.$$

{#eq-univ-score-stat}

The sign-flipping p value for a single specification is

$$p_{\text{flip}} = \frac{\sum_{r=1}^{B} \mathbb{I}\left(T^{(r)} \geq T^{(1)}\right)}{B}.$$

{#eq-flip-p}

The asymptotic validity of this meta-regression specialization is inherited from the general standardized sign-flipping score framework under the corresponding regularity conditions. As customary in conventional mixed-effects meta-analysis, the within-study sampling variances v_i are treated as known. We further assume that the meta-regression mean model is correctly specified, that the complete design matrix has full rank and fixed dimension as k increases, that the estimated working weights converge to finite and non-degenerate limits, and that the score contributions satisfy the regularity and Lindeberg conditions of De Santis et al. (2025a), excluding asymptotically dominating studies.

In particular, for the heterogeneity estimator used under the null, assume that

$$\hat{\tau}_0^2 \xrightarrow{P} \widetilde{\tau}^2$$

for some finite $\widetilde{\tau}^2 \geq 0$. For each study, continuity then gives

$$\hat{w}_{0i} = \frac{1}{v_i + \hat{\tau}_0^2} \xrightarrow{P} \frac{1}{v_i + \widetilde{\tau}^2} = \widetilde{w}_i.$$

Thus, estimation of τ^2 produces a convergent working-weight sequence of the type allowed by the general sign-flipping framework, provided the remaining regularity conditions hold. Importantly, $\widetilde{\tau}^2$ need not correspond to the true variance structure. De Santis et al. (2025a) show that, provided that the mean model is correctly specified and the required regularity conditions hold, the standardized sign-flip score test remains asymptotically valid under general variance misspecification.

Under these conditions, the difference between the flip-specific leading variance term used for standardization and the corresponding conditional variance vanishes in probability as $k \to \infty$. Therefore, the standardized study-level sign-flipping score test for $H_0 : \beta = \beta_0$ is asymptotically an α-level test.

The proposed test therefore inherits the robustness of the flipscores framework to misspecification of the study-level variance structure, provided that the meta-regression mean model and the other regularity conditions are correctly specified. This robustness does not extend to misspecification of the mean model or to violations of the assumed independence structure.

This test keeps the null model fixed and constructs the reference distribution by sign-flipping the study-level effective score contributions. For each tested null hypothesis, the constrained null meta-regression model is fitted once, and the B sign-flipping transformations are then applied directly to the resulting score contributions without re-estimating the regression coefficients or τ^2. This differs from `metafor::permutest()`, where the complete meta-regression model, including τ^2, is refitted for every permutation (Higgins & Thompson,

2004).

We explored this computational difference in the supplementary materials. For a meta-analysis with $k = 100$ studies and 3 predictors, `metafor::permutest()` takes roughly 14 seconds, compared with 0.3 seconds for the score sign-flipping approach. The score sign-flipping test is implemented in the `flipmeta` R package (https://github.com/filippogambarota/flipmeta). Hereafter, we refer to this extension of the flipscores framework to meta-analysis as flipmeta.

From `flipmeta` to the multiverse meta-analysis

The score sign-flipping test provides inference for a single meta-analytic specification. PIMA extends this logic to a multiverse by applying the same sign-flipping transformations across all admissible specifications and combining the resulting standardized score statistics across specifications (Girardi et al., 2024). We consider a fixed finite set of S meta-analytic specifications, indexed by $s = 1, \ldots, S$, while the asymptotic results refer to an increasing number of independent studies.

Specifications may differ in inclusion criteria, effect-size metric, moderator coding, heterogeneity estimator, or other analytical choices, provided that each specification can be represented within the independent mixed-effects meta-regression framework described above and tests a substantively compatible focal null hypothesis. Let β_s denote the focal coefficient in specification s and define

$$H_s : \beta_s = 0.$$

The global null hypothesis is the intersection of the specification-specific null hypotheses,

$$H_{\text{global}} = \bigcap_{s=1}^{S} H_s = \{\beta_s = 0 \text{ for all } s = 1, \ldots, S\}.$$

{#eq-global-null}

Rejecting this hypothesis implies that the focal effect is non-null in at least one specification, but does not identify which specification is responsible for the rejection (Girardi et

al., 2024).

To define common resampling transformations across specifications, let

$$\mathcal{I} = \{1, \ldots, K\}$$

denote the master set of independent studies occurring in the multiverse, and let $\mathcal{I}_s \subseteq \mathcal{I}$ denote the subset of studies included in specification s. For each transformation $r = 1, \ldots, B$, a single sign $g_i^{(r)} \in \{-1, +1\}$ is generated for each study $i \in \mathcal{I}$, with $g_i^{(1)} = 1$ for all i. The sign-flipping matrix used in specification s is then

$$\mathbf{F}_s^{(r)} = \operatorname{diag}\left\{ g_i^{(r)} : i \in \mathcal{I}_s \right\}.$$

{#eq-common-flips}

Hence, whenever a study occurs in multiple specifications, the same sign is applied to its score contribution in all of them. If a specification excludes a study, the corresponding element of the common sign vector is simply omitted. This is the construction used by PIMA for specifications involving different subsets of observations (Girardi et al., 2024). The common sign-flipping transformations preserve the joint resampling dependence among specification-level statistics induced by their shared study-level information.

For each specification s, its constrained null model is fitted and the standardized statistic $T_s^{(r)}$ is computed for each common transformation $r = 1, \ldots, B$. Inference across the multiverse can then be performed at two levels. First, a global test evaluates H_{global}. Following PIMA, the flipped statistics can in principle be combined using different non-decreasing functions; for post-selection inference, a particularly useful choice is the max-T statistic (Girardi et al., 2024),

$$M^{(r)} = \max_{s=1,\ldots,S} T_s^{(r)}.$$

{#eq-max-stat}

The corresponding global p value is

$$p_{\text{global}} = \frac{\sum_{r=1}^{B} \mathbb{I}\left(M^{(r)} \geq M^{(1)}\right)}{B}.$$

{#eq-global-p}

Rejecting the global null implies that the focal parameter is non-null in at least one specification, providing weak control of the family-wise error rate (Girardi et al., 2024).

The validity of this construction follows directly from PIMA once the specification-specific `flipmeta` tests satisfy the conditions described in the previous section. For a fixed finite number of specifications, if each specification satisfies the conditions required for the asymptotic validity of its standardized score test, the number of independent studies in each specification increases, and the same study-level signs are used across specifications, the resulting PIMMA global test is asymptotically an α-level test (Girardi et al., 2024). Specifications containing different subsets of studies are accommodated by restricting the common sign vector to the studies retained in each specification.

Second, selective inference can be performed for individual specifications. PIMA obtains multiplicity-adjusted p values through the max-T framework, which can be implemented using single-step or sequential step-down variants (Westfall & Troendle, 2008; Westfall & Young, 1992). Under the joint resampling results established for PIMA, these procedures provide strong control of the family-wise error rate, allowing inference on particular specifications selected after inspecting the multiverse (Girardi et al., 2024). This strong control follows from the joint resampling structure and the max-T procedure, rather than merely from accounting for dependence among the specification-level statistics.

Simulation study

We performed a simulation study to evaluate the inferential performance of the proposed `flipmeta` approach and compare it with standard Wald-type inference and the Knapp–Hartung method (Knapp & Hartung, 2003). Standard permutation-based procedures for meta-analysis (Follmann & Proschan, 1999; Higgins & Thompson, 2004) were not included because they

require repeated permutation of the model matrix and refitting of the complete meta-analytic model, resulting in a prohibitive computational cost.

Data were generated following the general simulation framework of Viechtbauer et al. (2015). We considered meta-analyses consisting of k independent studies, each comparing an experimental group (E) with a control group (C) on a quantitative outcome. For study $i = 1, \ldots, k$, observations in the two groups were assumed to follow normal distributions with means μ_i^E and μ_i^C and a common within-study standard deviation σ_i. The true standardized mean difference for study i was therefore

$$\theta_i = \frac{\mu_i^E - \mu_i^C}{\sigma_i}.$$

For each simulated study, we computed the observed standardized mean difference as

$$d_i = \frac{\bar{x}_i^E - \bar{x}_i^C}{s_i},$$

where $\bar{x}_i^E$ and $\bar{x}_i^C$ are the observed group means and s_i is the pooled standard deviation. The approximately unbiased estimator of θ_i was then obtained using the small-sample correction (Hedges, 1981),

$$y_i = \left(1 - \frac{3}{4\left(n_i^E + n_i^C\right) - 9}\right) d_i.$$

{#eq-g}

The corresponding sampling variance was estimated as

$$v_i = \frac{1}{n_i^E} + \frac{1}{n_i^C} + \frac{y_i^2}{2\left(n_i^E + n_i^C\right)}.$$

The vector of true study effects was generated according to the mixed-effects meta-regression model

$$\boldsymbol{\theta} = \mathbf{X}\boldsymbol{\beta} + \mathbf{Z}\boldsymbol{\gamma} + \mathbf{u},$$

where $\mathbf{X}$ contains the focal predictors, $\boldsymbol{\beta}$ their corresponding regression coefficients, $\mathbf{Z}$ contains nuisance predictors, $\boldsymbol{\gamma}$ their coefficients, and $\mathbf{u}$ is the vector of study-specific random effects. The random effects were independently generated as $u_i \sim \mathcal{N}(0, \tau^2)$.

Under the null hypothesis, the focal coefficients were set to $\boldsymbol{\beta} = \mathbf{0}$, allowing empirical Type I error rates to be evaluated. Under the alternative hypothesis, focal regression coefficients were set to $\beta \in \{0.1, 0.3, 0.5\}$, representing increasingly strong associations and allowing statistical power to be evaluated. The nuisance coefficients were similarly varied over $\gamma \in \{0.1, 0.3, 0.5\}$ under both the null and alternative hypotheses.

Focal and nuisance predictors were jointly generated from a multivariate normal distribution with mean zero and unit marginal variances. A compound-symmetry correlation structure with pairwise correlation $\rho = 0.5$ was used, so that focal and nuisance predictors were moderately correlated. This setting was chosen to represent the common situation in which inference on a focal moderator must account for correlated study-level covariates.

Residual heterogeneity was varied by setting $\tau^2 \in \{0, 0.01, 0.10, 0.50\}$. The condition $\tau^2 = 0$ corresponds to the absence of residual between-study heterogeneity, whereas the remaining values represent progressively larger amounts of heterogeneity on the standardized mean-difference scale. The number of studies was varied over $k \in \{10, 20, 40, 100\}$, covering meta-analyses ranging from relatively small to large. Study sample sizes were also allowed to vary across studies. For each study, equal expected group sizes were assumed, such that n_i^E and n_i^C were generated from the same negative-binomial distribution. Four average group sizes were considered, $n \in \{10, 30, 50, 80\}$, and the negative-binomial dispersion parameter was chosen so that the variance of the generated sample sizes was approximately twice their mean. This produces heterogeneous study sizes while retaining a prespecified average sample size, more closely reflecting the variation in study precision typically encountered in meta-analysis.

For each combination of simulation parameters, the resulting effect-size estimates y_i and

sampling variances v_i were analyzed using the competing inferential procedures. Performance under the null hypothesis was quantified by the empirical Type I error rate at the nominal significance level $\alpha = 0.05$, whereas performance under the alternative hypothesis was quantified by empirical statistical power.

The simulation results are summarized in Figure 1.

Overall, `flipmeta` maintained Type I error close to the nominal level across the conditions considered while retaining competitive power. Consistent with previous simulation studies (Viechtbauer et al., 2015), Wald-type inference showed inflated Type I error in some small-sample and misspecified-model conditions. For inference on a single meta-analysis, these simulations therefore show limited advantage of `flipmeta` over existing alternatives. Its main benefit emerges in the multiverse setting, where the common sign-flipping distribution can be used for both global inference and strong FWER control across correlated specifications. General procedures such as Bonferroni or Holm also control the FWER under arbitrary dependence, but do not exploit the dependence among tests and may consequently lose power when specifications are correlated.

In multivariate tests, and in particular in a multiverse analysis, the tests are likely to be correlated. Consider a multiverse analysis with two specifications based on the same set of k studies, differing only in whether an equal-effects model (i.e., fixing $\tau^2 = 0$) or a random-effects model is fitted. The score vectors, and consequently the sign-flipped test statistics, are likely to be strongly correlated, especially when τ^2 is close to zero. Considering (**?@eq-score-contribution-general**), for an intercept-only random-effects model under $H_0 : \mu = 0$, there are no nuisance predictors, $\tilde{x}_i = 1$, and $\hat{r}_{0i} = y_i$. The score contribution of the i-th study therefore simplifies to $v_i = \hat{w}_{0i} y_i$. Since y_i is identical across the two specifications, differences in the score contributions arise only from differences in the study weights, which in this example are mainly determined by the different values of τ^2. As $\hat{\tau}^2$ approaches zero, the random-effects weights approach the equal-effects weights, making the corresponding score vectors increasingly similar.

As a practical example, let's use the dataset from the meta-analysis by Credé et al. (2010)

about the relationship of class attendance with grades and student characteristics. The dataset is included in the `metafor` R package (Viechtbauer, 2010).

Code 1 shows substantial dependence among the test statistics, particularly for specifications differing only in the meta-analytic model. A Holm correction controls the FWER but does not exploit this dependence, potentially reducing power relative to max-T. The application below illustrates this comparison in a substantially larger multiverse.

Application to psychotherapy for depression

Dataset and original investigation

We illustrate PIMMA using the database assembled for the multiverse meta-analysis of psychological treatments for depression. The study was first described in a preregistered protocol by Plessen et al. (2022) and was subsequently completed and reported by Plessen et al. (2023). Its purpose was to determine whether conclusions about the efficacy of psychotherapy for depression were robust across the many defensible decisions available to meta-analysts. The authors searched PubMed, Embase, PsycINFO, and the Cochrane Central Register of Controlled Trials for randomized controlled trials published up to January 1, 2022. Studies comparing a psychological treatment with a control condition were eligible without restrictions on the type of psychotherapy, target population, treatment format, control condition, or definition of depression. The resulting database comprised 1,206 effect-size estimates from 415 studies and 71,454 participants. In the original investigation, combinations of inclusion criteria and statistical methods produced 4,281 meta-analyses. The results generally supported the efficacy of psychotherapy, while also showing that effect estimates varied systematically with choices such as the control condition, risk-of-bias restrictions, and adjustment for publication bias (Plessen et al., 2023).

The present application is not intended to reproduce the full multiverse or to provide a definitive assessment of the efficacy of psychotherapy for depression. Instead, we use a clinically more homogeneous subset of the database to illustrate post-selection inference across closely related meta-analytic specifications. We restricted the analysis to studies of CBT in adult populations. Before applying the specification-specific eligibility criteria, this application subset

contained 275 effect-size estimates from 86 independent studies. Studies contributed a median of 2 effect-size estimates (range: 1–20). Individual multiverse specifications then applied different restrictions on control condition, risk of bias, treatment format, and outcome rater, so the number of eligible effect sizes and studies varied across specifications. After applying these eligibility criteria and aggregating dependent effect sizes within study, each specification included between 6 and 86 independent studies.

Data preparation

The available standardized mean differences were expressed as Hedges' g, and their sampling variances were obtained from the corresponding standard errors. Records without an available effect-size estimate were excluded. Study identifiers and the categories used to define the multiverse—including treatment and control conditions, treatment format, target population, outcome rater, and risk of bias—were harmonized before applying the eligibility criteria. Risk of bias was represented by three categories: low risk, some concerns, and high risk.

Because a study could contribute multiple effect sizes, eligibility criteria were first applied separately within each specification. The retained effect sizes were then aggregated within study, so that each study contributed a single effect-size estimate and sampling variance to the corresponding meta-analysis. This aggregation step was necessary because the current PIMMA formulation assumes independent study-level contributions. Because the correlations among effect sizes from the same study were not available, rather than fixing the unknown dependence to a single value, we incorporated this uncertainty into the multiverse by varying the assumed within-study correlation across $\rho \in \{.30, .50, .70\}$. Rather than fixing this unknown quantity to a single value, we treated it as an additional multiverse decision and considered $\rho \in \{.30, .50, .70\}$. After aggregation, each specification therefore consisted of one effect-size estimate per retained study. The study, rather than the individual outcome or measurement occasion, was consequently the independent unit entering the meta-analysis and contributing to the study-level scores used by PIMMA. Importantly, aggregation was performed after applying the specification-specific eligibility criteria, because different specifications could retain different effect sizes from the same

study.

Multiverse specifications

All specifications estimated an intercept-only random-effects model, with the pooled Hedges' g as the parameter of interest. Following the framework proposed by Del Giudice and Gangestad (2021), we aimed to include analytical choices for which the alternative options could be regarded as approximately equally reasonable and defensible, rather than mechanically considering every technically possible analysis. The resulting multiverse therefore necessarily reflects researcher judgment. This is appropriate here because the application is intended to illustrate the inferential properties of PIMMA rather than to provide a definitive assessment of the efficacy of CBT for depression. Within this set of defensible alternatives, the multiverse varied four decisions governing which evidence was retained in each meta-analysis and two decisions concerning within- and between-study variance (Table 1). For each evidence-selection decision, the unrestricted level imposed no restriction on that characteristic and therefore retained all otherwise eligible categories. Crossing these decisions yielded $3 \times 3 \times 4 \times 3 \times 3 \times 2 = 648$ candidate specifications. To avoid performing meta-analyses on extremely sparse subsets, we required each specification to contain at least six independent studies. After applying this criterion, 438 specifications remained, representing 73 distinct subsets of studies and including between 6 and 86 studies per specification.

Results

The 438 retained specifications differed substantially in the amount of evidence included, ranging from 6 to 86 independent studies (median = 14, IQR = 9–27). The pooled Hedges' g estimates ranged from 0.36 to 1.30, with a median of 0.73 (IQR = 0.59–1.03). Despite this variability in magnitude, all pooled effect estimates were positive (Figure 3).

The global PIMMA test rejected the intersection null hypothesis that the average effect was zero across all 438 specifications (p = .0002). This provides evidence that the null hypothesis is false for at least one specification, but does not imply that all individual specifications can be rejected after accounting for multiplicity.

At the specification level, 426 of the 438 specifications (97.3%) had an unadjusted PIMMA p value below .05. After applying the single-step max-T correction, 378 specifications (86.3%) remained significant (Figure 2). Thus, 88.7% of the specifications that were significant before multiplicity correction remained rejectable after controlling the family-wise error rate across the multiverse.

The dependence among specification-level statistics was heterogeneous (Figure 4). Pairwise correlations had a median of 0.15 (IQR = 0.01–0.45), ranging from -0.04 to 1.00. This pattern is expected because some specifications differ substantially in the studies they retain, whereas others share the same or very similar study subsets and differ only in a small number of analytical decisions. A Holm correction applied to the same unadjusted PIMMA p values retained no specifications at the .05 level, compared with 378 (86.3%) under the single-step max-T procedure (Table 2). Because the same specification-level tests underlie both corrections, this comparison isolates the multiplicity adjustment: unlike Holm, max-T exploits the joint dependence structure while controlling the family-wise error rate.

Discussion

Methodologically, PIMMA addresses two complementary inferential levels. The `flipmeta` component provides score-based inference within each meta-analytic specification, while the joint PIMMA resampling scheme extends this inference across specifications, enabling both global testing and multiplicity-adjusted post-selection inference. This distinction is central to the proposed framework because valid inference within individual meta-analyses alone does not address the additional selection problem created by examining an entire multiverse.

The application illustrates how PIMMA separates two questions that are often conflated in multiverse analyses: whether there is evidence against the global null and which individual specifications remain rejectable after multiplicity correction. In the CBT example, all pooled estimates were positive but varied considerably in magnitude. The global null was rejected, while most, but not all, specifications remained significant after single-step max-T correction. The application also illustrates why dependence matters: specifications sharing studies or analytical

choices can generate correlated test statistics, and the common sign-flipping distribution allows max-T to exploit this structure. Accordingly, max-T retained substantially more specification-level rejections than Holm when both were applied to the same unadjusted PIMMA tests.

These results should therefore be interpreted primarily as a demonstration of the inferential framework rather than as a definitive reassessment of the clinical efficacy of CBT for depression.

Limitations and future perspectives

The main limitation of the present extension is the assumption that the meta-analytic dataset consists of independent studies contributing a single effect size to each specification. In practice, meta-analytic data often contain dependent effect sizes arising from multilevel or multivariate structures (Cheung, 2013; Van den Noortgate et al., 2013). The current implementation therefore requires dependent effects to be reduced to one study-level estimate before applying PIMMA, as in our application. Recent extensions of the flipscores framework have addressed nested and crossed random-effects structures by adapting the resampling scheme to the dependence among observations (Andreella et al., 2025; Vesely & Andreella, 2026). Extending PIMMA to these more complex meta-analytic structures is an important direction for future work.

A second limitation concerns the construction of the multiverse itself. PIMMA provides inference conditional on the specifications included, but does not determine which analytical choices should be considered reasonable or equally defensible. Their selection therefore remains a substantive and methodological judgment. In our application, we deliberately considered a limited set of plausible alternatives because the goal was methodological illustration rather than an exhaustive clinical assessment of CBT.

Finally, the proposed sign-flipping tests are asymptotically valid under the regularity conditions described above. Their finite-sample performance therefore remains important, particularly for meta-analyses with few studies or highly unbalanced study contributions. Our simulation study provides initial evidence of adequate Type I error control across the conditions considered, but further work is needed to evaluate performance under more complex forms of

heterogeneity, model misspecification, and dependence.

Conclusion

PIMMA extends post-selection inference to multiverse meta-analysis by combining study-level sign-flipping score tests with joint resampling across analytical specifications. It provides global and specification-specific inference while controlling the family-wise error rate and accounting for dependence among overlapping meta-analytic specifications. Simulations showed adequate Type I error control across the conditions considered, while the application illustrated how exploiting this dependence can yield substantially more specification-level rejections than a general-purpose correction such as Holm. PIMMA therefore complements descriptive multiverse meta-analysis by allowing researchers to assess not only how results vary across reasonable analytical choices, but also which inferential conclusions remain supportable after those choices are taken into account.

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Code 1 Correlations between test statistics in a multiverse meta-analysis

```r
library(metafor)
devtools::load_all()
set.seed(24617)

#library(flipmeta)

dat <- dat.crede2010
# calculating Fisher z transformed correlation coefficient
dat <- escalc("ZCOR", ri = ri, ni = ni, data = dat)
dat$id <- 1:nrow(dat)

# scenario 1, full data, random-effects
fit1 <- rma(yi, vi, data = dat, method = "REML")

# scenario 2, only journals articles
# (excluding dissertations and other sources), random-effects
fit2 <- rma(yi, vi, data = dat[dat$source == "journal", ])

# scenario 3, full data, equal-effects (tau2 = 0)
fit3 <- rma(yi, vi, data = dat, method = "EE")

fitl <- list(fit1, fit2, fit3)

# multiverse analysis, 1000 permutations
res <- flipmeta(fitl, B = 1000, id = "id", progress = FALSE)

# extracting the B x 3 matrix of permuted t statistics
# and calculating the correlations

R <- cor(res$Tspace)
round(R, 2)

# raw p values
res$summary_table$p
# maxT (step down) p values
p.adjust(res)$summary_table$p.adj
# bonferroni p values
stats::p.adjust(res$summary_table$p, method = "holm")
##               mod1:intrcpt mod2:intrcpt mod3:intrcpt
## mod1:intrcpt         1.00         0.83         0.61
## mod2:intrcpt         0.83         1.00         0.53
## mod3:intrcpt         0.61         0.53         1.00
## [1] 0.001 0.001 0.001
## [1] 0.001 0.001 0.001
## [1] 0.003 0.003 0.003
```

Table 1

Specification decisions defining the application multiverse.

Specification decision	Levels
Control condition	Waitlist; care as usual; unrestricted
Risk of bias	Low risk only; low risk or some concerns; unrestricted
Treatment format	Individual; group; guided self-help; unrestricted
Outcome rater	Clinician-rated; self-reported; unrestricted
Within-study correlation	ρ = .30, .50, or .70
Heterogeneity estimator	Restricted maximum likelihood; DerSimonian–Laird

Table 2
Specification-level inference across the CBT multiverse.

Method	Rejected specifications	Percentage (%)
Unadjusted PIMMA	426	97.3
Single-step maxT	378	86.3
Holm	0	0.0

Figure 1
Simulation results

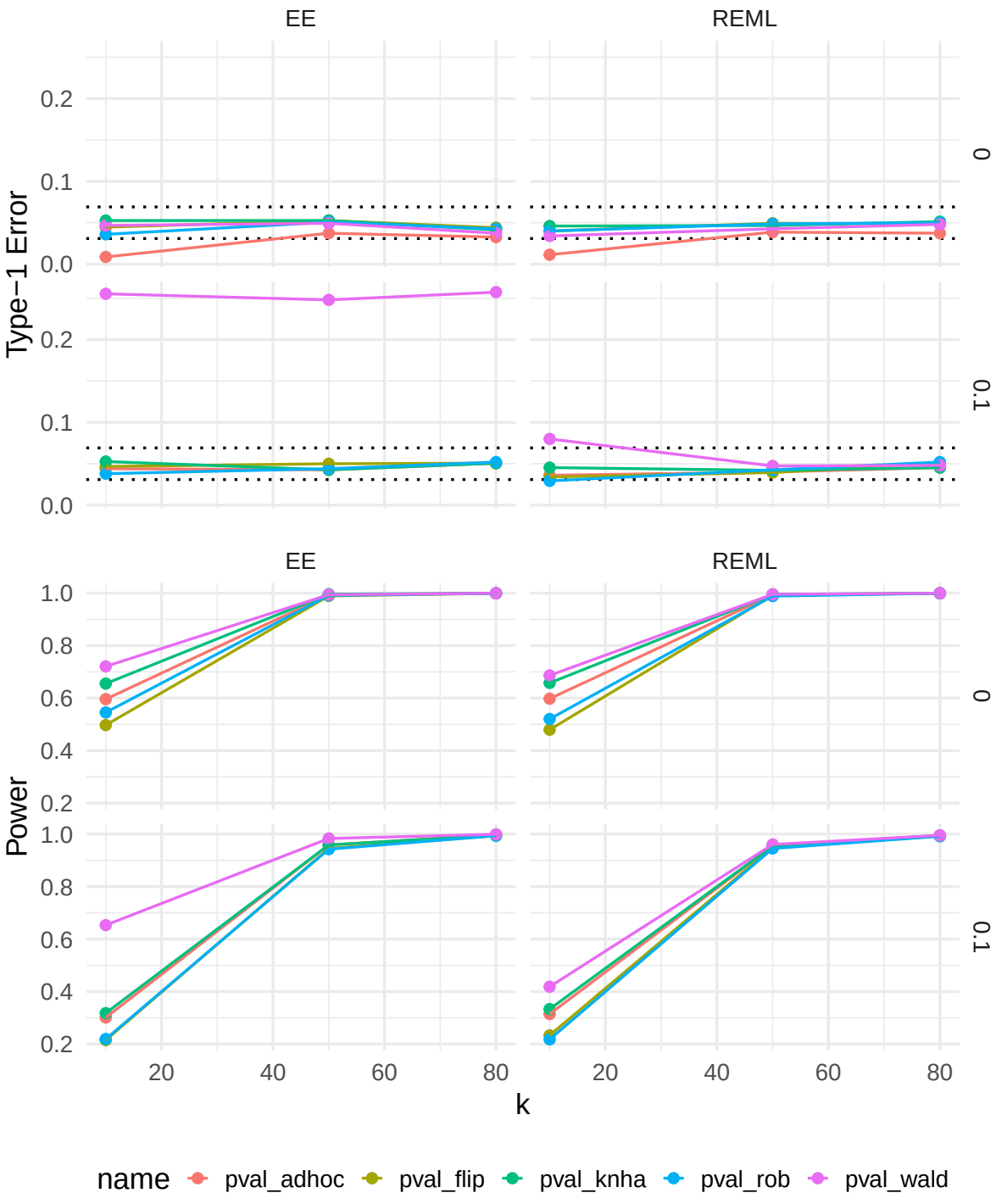


Figure 2

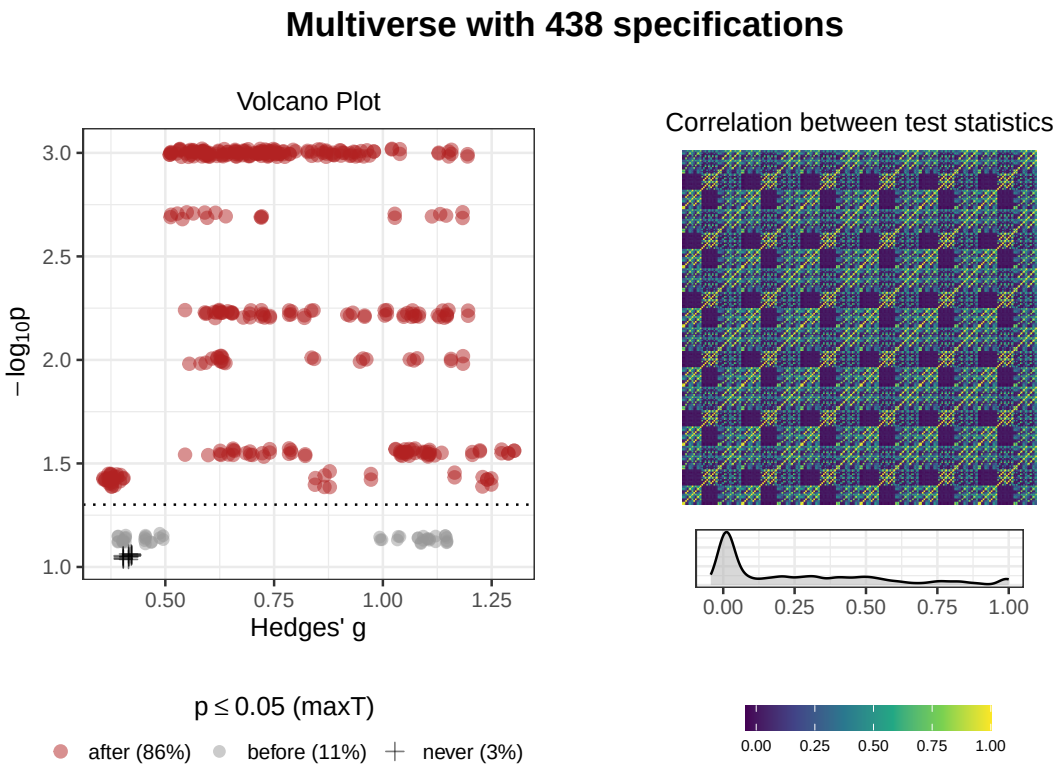

Multiverse with 438 specifications
Volcano Plot
Correlation between test statistics
3.0
2.5
2.0
1.5
1.0
$-\log_{10}p$
0.50
0.75
1.00
1.25
Hedges' g
0.00
0.25
0.50
0.75
1.00
p ≤ 0.05 (maxT)
after (86%)
before (11%)
never (3%)
0.00
0.25
0.50
0.75
1.00

Figure 3

Multiverse of meta-analytic estimates and analytical specifications. Specifications are ordered by the pooled Hedges' g. The upper panel shows the estimated effect for each specification and whether it was significant before and after single-step max-T correction. The dashed horizontal line indicates the median pooled Hedges' g across specifications, while the dotted line indicates the null value of zero. The lower panel shows the analytical decisions defining each specification.

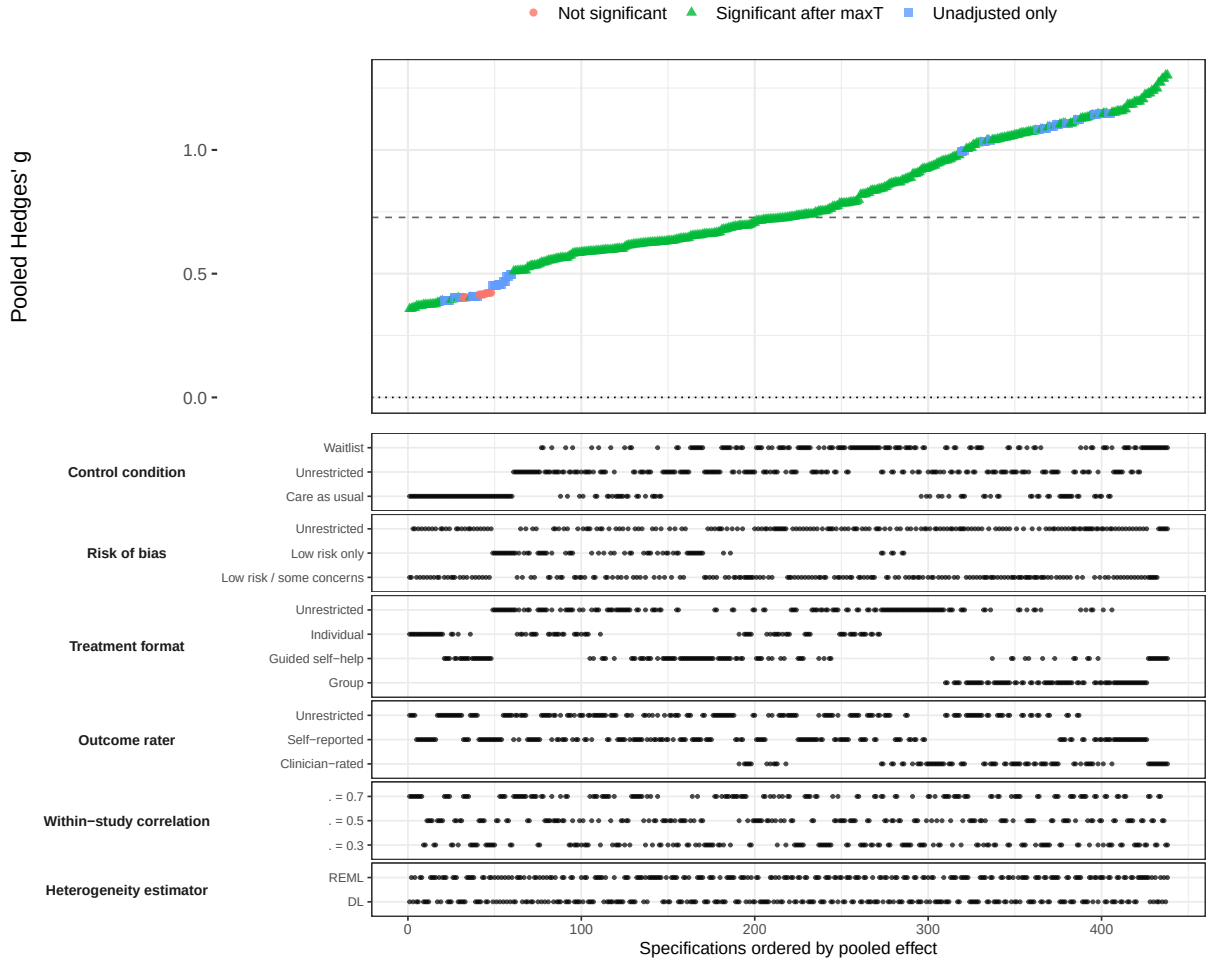


Figure 4

Dependence among specification-level test statistics. The left panel shows the pairwise correlation matrix of the joint sign-flipping test statistics across specifications. The right panel shows the distribution of the off-diagonal pairwise correlations.

